



ANTHROPOLOGY

Radiocarbon dates revise histories of Indigenous societies

New techniques challenge assumptions about Indigenous responses to European contact

LIZZIE WADE

Ten years ago, archaeologist Jennifer Birch thought she had a good handle on when and why Ontario's largest excavated Iroquois village, known as the Jean-Baptiste Lainé Site, was abandoned. Because archaeologists had found few European artifacts there, they figured people deserted the village around 1535, shortly after the first French explorers entered Canada's interior.

But Sturt Manning, a radiocarbon dating expert at Cornell University, saw a way to test that picture. He told Birch that with its wooden architecture and overlapping construction phases, the Jean-Baptiste Lainé Site would be a good candidate for a detailed chronology based on the latest innovations in radiocarbon dating. Birch, now at the University of Georgia (UGA), was stunned by the results: The site was, in fact, occupied between 1590 and 1615, roughly 75 years later than archaeologists thought and nearly a century after European contact. "I was in disbelief for 2 years," Birch remembers, and so were many of her academic peers. "We'd been off by two or three whole generations."

Now, the same technique is rewriting the history of other Indigenous sites along the East Coast of North America, as researchers reported during a session Birch co-organized at the Society for American Archaeology (SAA) meeting in Denver last week. Many of these new histories are challenging conventional wisdom about Indigenous persistence, or the lack thereof, in the face of European contact.

Traditionally, archaeologists have subdivided the history of eastern North American sites based on shifting ceramic styles, as well as the presence or absence of European artifacts. If a

site contains objects such as iron blades or glass beads, archaeologists include it in the so-called "historic" period, placing it earlier or later depending on the number of European artifacts. If a site lacks such items, researchers assume it was occupied in the precontact era.

"A lot of us are really frustrated by this arbitrary division of time," says Michelle Pigott, an archaeologist at the University of Tennessee, Knoxville. But in colonial eastern North America, it was often the best archaeologists could do. Radiocarbon dating during this time frame is particularly tricky.

The method works by measuring levels of carbon-14 in organic material such as wood, charcoal, and seeds. Living things absorb the radioactive isotope from the atmosphere, and it gradually decays after death. But turning carbon-14 levels into an age expressed in calendar years requires knowing how much radiocarbon was in the atmosphere when the wood or seed was alive. That baseline fluctuates over time—sometimes substantially, as it did during the 15th through the 17th centuries. For that period, radiocarbon dating can yield possible calendar years that are "wildly different but all completely plausible," says Jacob Holland-Lulewicz, an archaeologist at Pennsylvania State University, resulting in uncertainties of a century or more.

Now, dating experts such as Manning are sharpening the resolution with the help of sophisticated statistical models that incorporate multiple carbon dates from certain kinds of artifacts such as wooden posts or charcoal. Tree rings in those artifacts provide an independent check on the carbon dates, helping archaeologists distinguish which of the statistically pos-



Archaeologists are revisiting the histories of North American sites, such as those in Ocmulgee Mounds National Historical Park.

sible calendar years is most likely to be correct. The reduced uncertainty brings history into focus, down to just a few decades. “On a generation scale, you can tell a human story,” says Erik Marsh, a radiocarbon expert at Argentina’s National Scientific and Technical Research Council who applies similar methods in South America.

The new dates are helping answer questions tribal nations have about their own history. Victor Thompson, an archaeologist at UGA, is redating sites in Georgia’s Ocmulgee Mounds National Historical Park at the request of the Muscogee (Creek) Nation, whose citizens lived in the region until they were forcibly removed by the U.S. government to Oklahoma in the 19th century. Thompson presented unpublished data showing the Lamar site in Ocmulgee was very likely the Indigenous town known as Ichisi that was visited by Spanish explorer Hernando De Soto in 1540.

Other archaeologists had thought Ichisi was more likely another, nearby site that contains more European artifacts. But the new dates show that a lack of European trade goods doesn’t mean a site was precontact. “Just because European stuff exists, that doesn’t mean Indigenous people want it. They’re not passive recipients,” Holland-Lulewicz says.

Many Indigenous communities endured long after archaeologists

had assumed. Thompson’s new dates show that Muscogee ancestors lived at Lamar well into the 17th century, 100 years after De Soto’s visit. Pigott also presented preliminary radiocarbon data suggesting that villages belonging to the Joara society in North Carolina continued to be occupied long after their people destroyed the region’s first Spanish fort in 1568.

“People paint the picture that colonization was adopted abruptly—a one-night stay [by a European], and we changed our life completely. And that’s just not the case,” says Raelynn Butler, secretary of Culture and Humanities for the Muscogee (Creek) Nation and a discussant at the SAA session.

Improved radiocarbon dates can also free archaeologists from relying on European accounts to shed light on Indigenous people’s actions and decisions. At the session, Megan Anne Conger, an archaeologist who works in UGA’s radiocarbon dating lab, presented data on two sites where Indigenous people produced beaver pelts in southern Ontario. These sites were assumed to have been occupied after 1580, when the French formally established the North American fur trade. But radiocarbon dates show Indigenous people did not wait for a European decree to reinvent their economy. They were intensively hunting and processing beavers there decades earlier, driving supply before the French formalized their demand.

Harnessing the new dating techniques can be expensive. “Quality control is everything,” Manning says. Birch adds that unlike artifact-based chronologies, which are often generalized over entire regions, each carbon-based “microhistory” is unique to a site, and must be “scaffolded together” to construct a larger historical narrative.

Butler emphasized that the highly technical nature of the research can create new barriers between archaeologists and Indigenous communities. “It shouldn’t take a tribal historic preservation officer luckily reading a paper to make that connection with the researcher, to share those histories, to strengthen what we know,” she said. She also called on archaeologists to connect their new radiocarbon chronologies all the way up to tribal nations in the present day. “Why is it that this [contact] time period is so critical and important?” she said to close the session. “And not necessarily what happened to our people after that?” □

TRUMP TRACKER

HOLD THAT THOUGHT After public and scientific criticism, President Donald Trump’s administration on 24 April rapidly reversed course on a landmark study of women’s health that the National Institutes of Health (NIH) planned to gut and dropped a key component of a controversial new probe of autism’s cause. Last week, the heads of regional centers integral to NIH’s Women’s Health Initiative said they were told the centers’ contracts would end in September. Following an outcry led by senators and celebrity Maria Shriver, a spokesperson for the Department of Health and Human Services (HHS) called that a mistake and said: “We are now working to fully restore funding to these essential research efforts.” Separately, NIH Director Jayanta “Jay” Bhattacharya indicated in public comments last week that his agency would set up a registry of individuals with autism, amassing data from sources including private entities, as part of HHS Secretary Robert F. Kennedy Jr.’s project to unravel the condition’s causes. There was immediate protest from the autism community and privacy advocates. Days later, HHS declared: “We are not creating an autism registry.”

NEW HEADS NIH last week named acting directors for six of its institutes, including the agency’s second largest overall, the National Institute of Allergy and Infectious Diseases (NIAID). Influenza researcher Jeffery Taubenberger will lead NIAID after the Trump administration forced out its former director, as well as the other institute heads. Taubenberger’s team famously sequenced the virus that caused the 1918 flu pandemic, extracting samples from autopsies of soldiers and an Inuit woman buried in the Alaskan permafrost. All six appointees are experienced NIH employees, and some—including the new acting directors of the Institute on Minority Health and Health Disparities and the Institute of Child Health and Human Development—have already been deputy directors.

REPORT AUTHORS FIRED The Trump administration this week fired the volunteer authors working on the sixth iteration of the U.S. National Climate Assessment, the congressionally mandated report, due later this decade, that guides businesses and states adapting to global warming. The move was not unexpected: Last month, the White House fired technical staff supporting the report. Scientists fear the administration will now try to produce a report downplaying the risks of climate change.